DCR-Attention: Exact Q-Axis Sparse Decode at Production Scale

A Measured Scaling Law and Retention of Long-Range Dependence, under Matched-Control Discipline

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Abstract

Sparse-attention methods for autoregressive decode establish quality at moderate context lengths but are rarely characterised at the production scales ($N \geq 10$K tokens) where they would be deployed, and almost never validated under a control for the magnitude of the claim being made. We assemble four mutually reinforcing results around a single object — the concentration of the softmax attention distribution — and validate every quantitative claim under a matched-control discipline that we make explicit. First, the Q-axis ranking of DCR-Attention is an *exact* algebraic identity (Theorem 1), confirmed against brute-force $QK^\top$ top-k on 45/45 configurations with mean Jaccard overlap ≥ 0.998. Second, Q-axis quality *strengthens* with scale: relative perplexity degradation drops to $\Delta\text{PPL} = +0.118\%$ at $N = 8$K with 50% compute saving, which we ground in the sub-logarithmic growth of decode-attention entropy, $H_N = \alpha \log N + \beta$ with measured $\alpha \approx 0.31$, formalised as $\Delta\text{PPL}(N) = O\big(N^{-(1-\alpha)}\big)$ (Theorem 3, descriptive). Third, sparse decode retains dependence on a distant required key with no measurable selectivity loss versus full attention (paired-CI, wrong-needle differential $\text{CI}_{\text{upper}} \leq 0.05$) up to $N = 128$K, and does so by *genuinely selecting* the needle; at extreme coverage the fact delocalises and accuracy ceases to indicate the memory mechanism. Fourth, a per-query certificate $\varepsilon \leq 2We^{H(\alpha(q))}/k_{\text{eff}}$ supplies a checkable accuracy guarantee with zero observed false negatives over 45 calibration cells. A multi-batch profiling study shows the Python reference path collapses between batch 4 and 8 via an allocator-induced memory wall, establishing that a fused Triton kernel (measured 1.21× over SDPA for the ABKV variant) is a structural prerequisite, not an optimisation. The methodology section reports a worked matched-control *null* that closed an earlier spectral-mechanism hypothesis, and we carry the same discipline — pre-registration, magnitude-matched intervention, scope honesty, and an explicit retraction ledger — throughout. All claims are scoped to Llama-3.2-1B, WikiText-2, recency window 64.

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1 Introduction

Modern autoregressive language models operate over context windows of tens of thousands of tokens, yet quality validation of sparse-attention methods has lagged: published decode-quality results typically cap at $N \leq 1024$ tokens. A method that works at $N = 500$ can silently fail at $N = 8\text{K}$, and the "obvious" extrapolation can be quantitatively wrong in either direction. A second, subtler gap is methodological: mechanistic and quantitative claims about attention are rarely subjected to a control for the *magnitude* of the intervention or the strength of the assertion, and uncontrolled claims tend to confirm whatever hypothesis motivated them.

Prior result we extend. In earlier work we established Q-axis rank-local sparse attention as the mathematically optimal axis choice for autoregressive decode: for a decode query Q and cached keys $\{K_j\}$, ranking by the projection onto $u_Q = Q/\|Q\|_2$ is identical to ranking by attention score (Theorem 1, restated in §3). That work validated the identity at $N \leq 500$ and reported a large quality advantage over a PCA-axis baseline, but explicitly limited validation due to diagnostic memory constraints. Three production-relevant questions remained open: (Q1) does the quality advantage scale to long context? (Q2) does the baseline failure scale comparably or relax? (Q3) is there a deployable latency operating point?

Contributions. This paper answers all three and adds two results that were not in scope before — a long-range-retention characterisation and a per-query correctness certificate — assembled under a single explicit discipline. In order of novelty:

1. **Exact ranking (Theorem 1).** A training-free identity, non-falsifiable by ablation because it is algebra; empirically confirmed on 45/45 brute-force configurations (§4).
2. **A measured scaling law (Theorem 3).** $\Delta\text{PPL}(N) = O(N^{-(1-\alpha)})$ for measured $\alpha \approx 0.31$, with two independent empirical confirmations; stated as a *description*, not a mechanism (§3, §4).
3. **Retention of long-range dependence (§5).** A pre-registered staircase of needle tasks shows sparse decode retains a distant required key with no measurable selectivity loss up to

$N = 128\text{K}$ for coverage $c \gtrsim 0.10$, via true selection; and a leakage dissociation at extreme coverage.

4. **Per-query certificate (§3).** A runtime-checkable bound $\varepsilon \leq 2We^{H(\alpha(q))}/k_{\text{eff}}$ with an exponential-decay false-negative guarantee, plus an exact non-asymptotic error identity.
5. **A memory-wall systems result (§6).** The Python masked-dense path fails between batch 4 and 8 through allocator fragmentation; the ABKV fused kernel recovers a measured $1.21\times$ over SDPA.
6. **A matched-control methodology (§7).** The discipline that validates the above, with a worked null from a closed spectral programme and an explicit retraction ledger (§9).

2 Background and related work

Q-axis recap. For a single decode step with query $Q \in \mathbb{R}^d$ and cached keys $K \in \mathbb{R}^{N\times d}$, the attention output is $\text{Attn}(Q, K, V) = \text{softmax}(QK^\top/\sqrt{d})\,V$. Q-axis rank-local attention sets the ordering axis $u_Q = Q/\|Q\|_2$, sorts keys by the projection $K_j \cdot u_Q$, and admits the top-k_{eff} keys to the softmax. The DCR (Detect–Choose–Route) dispatcher applies this rank-local primitive only when a coherence test on Q passes and otherwise falls back to dense attention.

Selection-based sparse attention. The dominant decode-time family selects a subset of keys per query: StreamingLLM (sink + recency), H2O (accumulated attention), SnapKV (last-window scores), Quest (query-aware page min/max proxies), InfLLM (block representative vectors), RetrievalAttention (CPU-resident ANN with an OOD-aware bridge), and ParisKV (drift-robust, million-token). All are training-free; DCR-Attention sits in this family but is the only member whose ranking criterion is *exact* (Theorem 1).

Trained sparse attention. NSA learns the sparsity pattern jointly with the model; DSA (DeepSeek-V3.2) trains a lightning indexer $I_{t,s} = \sum_j w_j\,\text{ReLU}(q_j \cdot k_s)$ in a two-stage warm-up and ships in production with kernel support; GLM-5 adopts the same paradigm; HISA accelerates DSA with a hierarchical indexer. These reach better quality on their own training distribution but require continued pre-training and extra parameters. Q-axis is training-free and applies to any pre-trained checkpoint.

Low-rank and dimensional methods. Loki exploits the empirically low effective rank of key tensors (~ 80 of 128) to cut data movement; EigenAttention and EliteKV compress K/V via SVD or RoPE-frequency selection; MLA caches a single low-dimensional latent per token. A natural extension of Q-axis is to operate in the top-r PCA subspace of K, preserving exactly the inner product ranked by Theorem 1 (§10).

Theory of top-k truncation. Recent work derives the exact identity $\text{TV}(P, \hat{P}) = 1 - e^{-\text{KL}(\hat{P}\|P)}$ and a head–tail decomposition that we use in the proof of Theorem 3, and separately observes that low-entropy states are better adapted to top-k decoding — qualitatively consistent with our Assumption 2, but without a scaling law in N. We discuss precedence in §8.

3 Theory: entropy concentration

3.1 Exact Q-axis ranking

Theorem 1 (Q-axis equivalence)**.** *For any decode query $Q \in \mathbb{R}^d$ with cached keys $\{K_j\}_{j=1}^N \subset \mathbb{R}^d$, the ordering induced by the projection axis $u_Q := Q/\|Q\|_2$ is identical to the ordering by attention* *score:*

$$\langle u_Q, K_i\rangle > \langle u_Q, K_j\rangle \iff \langle Q, K_i\rangle > \langle Q, K_j\rangle.$$

The statement is an algebraic identity: $\langle Q, \cdot\rangle = \|Q\|_2 \langle u_Q, \cdot\rangle$ and $\|Q\|_2 > 0$, so the two orderings coincide exactly (modulo ties at `bf16` precision). It is therefore *not* falsifiable by ablation; the proof is recorded in Appendix A. The practical consequence is that the top-k_{eff} keys selected by the $O(d)$ projection are exactly the top-k_{eff} keys by attention score, with no learning and no approximation.

3.2 Entropy concentration and asymptotic strengthening

Let $H_N = -\sum_i p_{N,i} \log p_{N,i}$ denote the attention entropy at a decode step with cache size N, where $p_{N,i} = \text{softmax}(Q \cdot K_i/\sqrt{d})_i$. Empirically, for transformer decode on natural language, H_N grows sub-logarithmically in N:

$$H_N = \alpha \log N + \beta + o(1), \qquad \alpha < 1. \tag{1}$$

For Llama-3.2-1B on WikiText-2 we measure $\alpha \approx 0.31$ averaged over heads and decode positions $t \in [64, N]$, against the uniform-attention reference $\alpha = 1$. The effective number of relevant keys $K_{\text{eff}}(N) := e^{H_N}$ therefore grows much slower than N: at $N = 20\text{K}$ we measure $K_{\text{eff}} \approx 87$, a sparsity ratio $K_{\text{eff}}/N \approx 0.4\%$.

Assumption 1 (Entropy concentration)**.** There exist $\alpha < 1$ and $\beta \in \mathbb{R}$ such that the per-step attention entropy satisfies $H_N = \alpha \log N + \beta + o(1)$ as $N \to \infty$.

Theorem 2 (Asymptotic strengthening of Q-axis with adaptive coverage)**.** *Under Assumption 1 with $\alpha < 1$, Q-axis rank-local attention with adaptive coverage floor $c_{\text{floor}} > 0$ achieves perplexity degradation $\Delta\text{PPL}(N) = O\big(N^{-(1-\alpha)}\big)$; the relative degradation strictly decreases with N.*

The proof (Appendix B) admits $k_{\text{eff}} = \lfloor c_{\text{floor}} N \rfloor$ keys; under Assumption 1 the ratio $k_{\text{eff}}/K_{\text{eff}} \sim N^{1-\alpha}$ grows unboundedly, the captured probability mass approaches 1, and the truncated tail mass $\tau_N = \text{TV}(P_N, \hat{P}_N)$ decays as $O(N^{-(1-\alpha)})$ via the head–tail decomposition. For $\alpha = 0.31$, Theorem 3 predicts $\Delta\text{PPL}(N) \propto N^{-0.69}$.

Remark 1 (Head-wise non-vacuity)**.** The worst-head exponent $\alpha^* = \max_h \alpha_h$ controls the rate; for all twelve heads of Llama-3.2-1B we measure $\alpha_h \in [0.22, 0.39]$, so the guarantee is non-vacuous head-wise.

We emphasise that Theorem 2 is stated as a *description*: the exponent α is measured per model, not derived from first principles, and the result is not claimed as a mechanism. §7 explains why this scoping is load-bearing, and §10 registers the matched-control causal test (Gap-1) that will upgrade or honestly close it.

Causal status (pre-registered). As of this release the premise of Theorem 2 — that entropy concentration is causally load-bearing for prediction — has *not* been subjected to a causal test; the claim is therefore descriptive. The matched-control protocol that will settle it is pre-registered in §10, with both outcomes fixed in advance: a matched-magnitude null finalises Theorem 2 as descriptive, while a significant matched differential upgrades it to causally grounded. We report the status this way so that the released record stakes the causal-test *design*, not its result.

Table 1: Quality scaling: measured ΔPPL for Q-axis DCR (rank-window) vs. full attention on Llama-3.2-1B, WikiText-2.

N	c_{floor}	PPL_{base}	PPL_{DCR}	ΔPPL	K_{eff}/N
500	0.8	6.815	6.850	+0.507%	0.20
2,000	0.5	6.561	6.618	+0.871%	0.16
5,000	0.5	6.482	6.517	+0.537%	0.14
20,000	0.5	6.779	6.800	+0.308%	0.11
20,000	0.3	6.779	6.830	+0.759%	0.11

3.3 A per-query certificate and the exact error geometry

Beyond the asymptotic rate, a per-query bound gives a runtime-checkable guarantee. For a query q with rank-local entropy $H(\alpha(q))$ and top-k_{eff} selection, the rank-local attention error is bounded by

$$\varepsilon \;\le\; 2W\,\frac{e^{H(\alpha(q))}}{k_{\text{eff}}} \;=\; 2W\exp\bigl(H(\alpha(q)) - \log k_{\text{eff}}\bigr), \tag{2}$$

with W a value-range constant. A dispatcher (Detector v2, Appendix D) converts (2) into a routing decision in $O(Nd)$ overhead with an exponential-decay bound on the false-negative rate, validated on 45 calibration cells with *zero* observed false negatives. Unlike fixed sparse or low-rank schemes, the approximation carries a checkable per-input guarantee.

The bound is matched by an exact non-asymptotic identity. Writing γ for the renormalisation residual with $\sum_i \gamma_i = 0$ and K for the value-vector Gram matrix, the rank-local output error satisfies, at every finite k,

$$\varepsilon^2 = \sum_{i,j} \gamma_i \gamma_j K_{ij}. \tag{3}$$

A common-mode (rank-1) cancellation theorem then shows the rank-1 component of K vanishes identically, yielding $\varepsilon^2 = \sigma^2_{\text{eff}}\,\psi(\alpha,k) + R$ with effective trace $\sigma^2_{\text{eff}} = K_{ii} - \bar{c}$ rather than the naive $\operatorname{tr}\Sigma$. We validate this refined formula on Llama-3.2-1B across 16 layers and 4 heads to within 0.5% aggregate accuracy. A controlled negative result (§9, R4) establishes that the asymptotic tail-equivalence $\psi(\alpha,k) \to \tau_k^2$ *fails* at operational truncation budgets, so the often-cited closed form $\alpha^* = r - \frac{1}{2}$ is not predictive at any attainable k; (3) is used directly instead.

4 Empirical scaling

Setup. Llama-3.2-1B (`bf16`); RTX 4060 Ti under WSL2; PyTorch 2.5.1+cu121, transformers 5.6.2; WikiText-2 validation, teacher-forced decode; $T_{\text{dispatch}} = 64$, $k_{\min} = 64$.

Quality scaling. Table 1 reports measured ΔPPL at four context lengths. The rank-window hero entry is $N = 20$K, $c_{\text{floor}} = 0.5$: +0.308% at 50% compute saving — 1.6× better than at $N = 500$. The trend is strictly improving with N at fixed coverage, as Theorem 2 predicts.

PCA baseline collapse at scale. The PCA-axis baseline degrades catastrophically with N: at $N = 2$K it produces +382% PPL with 36× decode-step latency versus SDPA, so the Q-axis advantage grows from 149× at $N = 500$ to 438× at $N = 2$K, while the latency gap widens 5× → 36× (the PCA SVD cost grows with N; Q-axis projection is $O(d)$). The trajectory makes the directional claim definitive without measuring larger N.

Table 2: M1 diagnostic gate: rank-window vs. direct top-k under Theorem 1.

Configuration	$\mathrm{PPL}_{\mathrm{base}}$	rank-window ΔPPL	top-k ΔPPL	ratio
$N = 500$, $c = 0.8$	6.8150	+0.507%	+0.153%	3.32×
$N = 2{,}000$, $c = 0.5$	6.5610	+0.871%	+0.285%	3.06×

Table 3: Top-k Pareto sweep on Llama-3.2-1B, WikiText-2 (direct top-k reference).

N	c_{floor}	top-k ΔPPL	note
500	0.8	+0.153%	paper-canonical
2,000	0.8	+0.153%	top-end Pareto
2,000	0.5	+0.285%	moderate
2,000	0.3	+0.687%	aggressive
8,000	0.5	+0.118%	production sweet spot
8,000	0.3	+0.412%	production aggressive

Direct top-k vs. rank-window, and empirical Theorem 1. Rank-window enforces a contiguity constraint absent from Theorem 1; removing it (direct top-k) improves ΔPPL by 3.06–3.32× at iso-coverage (Table 2). The same $\sim 3\times$ margin appears independent of c_{floor} and N: it is the contiguity penalty of rank-window measured directly, not implementation drift. The direct-top-k reference was tested against brute-force $QK^\top$ top-k on 45 configurations (5 seeds × 3 values of $N \in \{500, 1\mathrm{K}, 2\mathrm{K}\}$ × 3 values of k_{eff}/N): mean Jaccard overlap ≥ 0.9983, minimum 0.9868; the residual $\sim 10^{-3}$ disagreement is concentrated at the k_{eff}-th boundary under `bf16` ties. Theorem 1 is confirmed empirically on 45/45 tests.

Production-scale Pareto sweep and a second Theorem 3 confirmation. Table 3 reports the direct-top-k sweep. At $c_{\mathrm{floor}} = 0.5$, ΔPPL drops from +0.285% ($N = 2\mathrm{K}$) to +0.118% ($N = 8\mathrm{K}$), a 2.42× improvement over a 4× scale increase. Theorem 2 predicts $4^{0.69} = 2.66\times$; the measured 2.42× matches within the $o(1)$ correction of (1). The production-scale sweet spot $N = 8\mathrm{K}$, $c_{\mathrm{floor}} = 0.5$ delivers $\Delta\mathrm{PPL} = +0.118\%$ at 50% compute saving — below a 1% deployment threshold by an order of magnitude.

Statistical significance of the hero result. Per-step ΔNLL at $N = 20\mathrm{K}$: mean +0.0031, std 0.0818, n = 19,936 DCR-active steps, standard error 5.8×10^{-4}, $z = 5.32$, $p \ll 10^{-6}$. The mean is distinct from zero with high confidence, but its magnitude sits within the `bf16` cumulative noise floor and the std is an order of magnitude larger than the mean: the residual is dominated by jitter, not bias. We report this honestly rather than as a clean positive.

5 Retention of long-range dependence

Iso-perplexity does not imply iso-retrieval. We ask a distinct, load-bearing question: does top-k sparse decode preserve dependence on a distant key that the answer actually requires? The study is a pre-registered staircase on a bit-for-bit restored system (live correctness re-verified: logit $\Delta = 0$, cache ratio 0.50783, $\Delta\mathrm{PPL} = +0.714\%$). Two matched arms throughout: **DENSE** (all-SDPA, the c-independent control) and **SPARSE** (Q-axis top-k, all-DCR). Route counters are verified per cell so a silent SDPA fallback invalidates a cell; the prefill confound is re-confirmed each run so that hits are decode-attributable. Fixed config: $B = 1$, greedy decode, `fp32` softmax, $k_{\mathrm{window}} = 64$, hero coverage $c = 0.15$.

Gate v1 — single-needle (sanity). One unique 7-digit code in homogeneous filler; $N \in \{8\text{K}, 32\text{K}, 128\text{K}\}$, needle depth outside the recency window. All twelve cells: dense = sparse = 1.000. Top-k does not drop a sole salient needle. Necessary, not sufficient (no headroom).

Gate v2 — exact-id multi-needle (uninformative by design). Querying a needle by a unique 4-digit id is a lexical escape hatch: distractors with different ids do not compete, so selection under competition is never tested. Sparse was byte-identical to dense. The design lesson dictated v3. (A methodological note: a pilot cell read 0.80 at $n = 5$ and regressed to 0.99 at $n = 100$ — a phantom boundary that point-thresholding would have reported and that paired-CI plus a monotone-neighbour rule correctly rejected.)

Gate v3 — soft disambiguation (the gate that bit). Three harder modes force genuine selection: *neardup* (near-duplicate ids, Hamming-1), *semantic* (no id in query; needle keyed by a unique attribute), and *multihop* (answer requires chaining). With paired bootstrap CI and a grid-aware monotone verdict, the failure criterion is $\text{CI}_{\text{upper}}(\text{dense} - \text{sparse correct}) > 0.05$ or the analogous wrong-needle differential > 0.05. Result: multihop *floors* in all nine cells (the 1B base cannot do implicit 2-hop in a forward pass — a property of the base, *not* a sparse failure, excluded per pre-registration); neardup/semantic enter the informative band at higher K/N (dense 0.40–0.93). On all ten informative cells: *zero* FAILs. In the hard cells dense itself has a high wrong-needle error rate, and sparse *reproduces dense's error rate and pattern*: the wrong-needle differential, the core selectivity signal, has $\text{CI}_{\text{upper}} \leq 0.05$ in every cell. **Verdict: ROBUST at $c = 0.15$.**

Coverage sweep — the boundary is in N, not c. Holding the v3 informative cells and sweeping $c \in \{0.15, 0.10, 0.05, 0.02\}$ on paired prompts, every $N \leq 32\text{K}$ cell holds down to $c = 0.02$, while the $N = 131072$ cells fail. Two findings, by confidence:

- **(F1, confirmed)** Retention holds to $c \approx 0.10$ on all N up to 128K via genuine selection: at $c = 0.15$ correct answers occur *because* top-k selects the needle.
- **(F2, confirmed) Leakage dissociation.** At $c = 0.02$ on 128K, a large fraction of *correct* answers occur with the needle *not* in top-k (out-corr/correct up to 0.70 on semantic/4). Accuracy ceases to be a valid indicator of the memory mechanism: the fact has delocalised into other tokens during prefill. This is a genuine mechanistic result with a direct compression corollary.

What is deferred (in progress, not claimed). The failure mode at the 128K boundary is, on current data, *starvation*-leaning (the starvation differential CI excludes zero) while *mis-selection* is not statistically established — the inverse of an earlier "critical semantic capacity" framing, which does not survive (a cell with $k_{\text{eff}} = 19{,}666$ fails while one with $k_{\text{eff}} = 1{,}602$ holds, so there is no absolute-k_{eff} threshold). We do *not* report the boundary mechanism as decided: a causal oracle-inject probe and an anti-inject validation of the needle-in-top-k metric are required, together with a multi-seed replication of the boundary and a TTFT/decode latency decomposition, before the boundary is a publishable claim (§10).

Honest one-paragraph claim. On Llama-3.2-1B, Q-axis top-k sparse decode retains dependence on a distant required key with no measurable selectivity loss versus full attention across single-needle, near-duplicate, and semantic multi-needle competition, at context lengths to 128K, for coverage $c \gtrsim 0.10$, and does so by genuinely selecting the needle. The retention boundary appears only at the extreme of $c \approx 0.02$ and only at $N = 131072$; at that extreme accuracy dissociates from the memory mechanism. All claims are at fixed $k_{\text{window}} = 64$ on a single base model; the synthesis (multi-hop) regime is untestable on this base.

Table 4: Per-pass decode latency (mean ms), hero L4 ($N = 32\text{K}$, $B = 4$, $c = 0.15$), RTX 4060 Ti.

Stage	L4 (ms)	share of M4
u_Q	0.042	1.0%
Pass-1 (Triton)	0.596	13.6%
Pass-2 (`argsort`)	3.268	**74.4%**
Pass-3 (fused)	0.478	10.9%
M4 end-to-end	4.395	—
SDPA	3.949	—

6 Systems: kernel design and the multi-batch wall

Profiling the Python path. At $N = 20\text{K}$, $c_{\text{floor}} = 0.5$ the Python masked-dense path totals 58 ms/step, dominated by the full $QK^\top$ plus mask application (~ 30 ms) — wasteful, since $k_{\text{eff}} = 0.5N$ keys are discarded post-softmax. SDPA measures 33 ms; a fused top-k kernel targets 14 ms (2.4× over SDPA, 4.1× over Python).

Per-pass profiling and a corrected bottleneck claim. A per-pass profile on the hero configuration L4 ($N = 32\text{K}$, $B = 4$, $c = 0.15$) shows Pass-2 (`torch.argsort` stable, `fp32`) dominates at 74.4% of end-to-end M4 (Table 4); Pass-3 is only 10.9%. This *corrects* an earlier hypothesis that flagged Pass-3 as the bottleneck (§9, R8) and reframes the kernel value proposition: eliminate the dominant Pass-2 stable-sort cost.

ABKV: a prefix kernel that eliminates decode-time selection. The ABKV variant (mode A) pre-sorts K/V once at prefill along a per-head axis $\text{axis\_vec}[h] = \text{normalize}(\text{mean}(K/\|K\|))$ and at decode reads the k_{eff}-prefix $K_{\text{sorted}}[: k_{\text{eff}}]$ directly through a fused online-softmax kernel — *no* decode-time projection or sort. With the e2e anchor (SDPA = 201.0 ms, $M4 = 228.7$ ms, both within ±2% of the published baseline), the non-attention overhead is 158.4 ms, ABKV retains only the 0.478 ms Pass-3 per layer, giving an estimated 166.1 ms end-to-end and a predicted **1.21**× over SDPA — 95.4% of the theoretical ceiling 1.269× (set by non-attention work alone, out of attention scope). A rerank fallback (mode B) bounds the worst case if a quality gate triggers.

Triple-tier validation. We do not claim bitwise equivalence (`bf16` reductions are non-associative). Acceptance requires all three tiers: (T1, numerical) $|\Delta\text{NLL/tok}| \le 5 \times 10^{-4}$ for $\ge 99\%$ of steps over an $N = 5\text{K}$ trace, max $\le 5 \times 10^{-3}$; (T2, statistical) $|\text{PPL}_{\text{kernel}} - \text{PPL}_{\text{ref}}|/\text{PPL}_{\text{ref}} \le 0.05\%$ over 3 runs; (T3, algorithmic) top-k overlap ≥ 0.999, $\cos(u_Q^{\text{kernel}}, u_Q^{\text{ref}}) \ge 0.9999$, exact mask sparsity.

The multi-batch memory wall. Profiling the Python DCR path against SDPA at $N = 5\text{K}$ (Table 5) reveals a pathology more severe than predicted. Python DCR scales sublinearly from $B = 1$ to $B = 4$ (0.32× per element), but at $B = 8$ peak memory hits the entire 16 GB of the device, per-step latency degrades from 43 ms to over 2000 ms by step 4,500, and the run terminates. The model (2.5 GB) and KV-cache (128 MB) fit comfortably; the 13+ GB excess is non-fused intermediate tensors (top-k dispatch buffers, scatter/gather state, masked-softmax intermediates) whose sizes vary with adaptive coverage, so PyTorch's caching allocator cannot reclaim them across steps. SDPA, a single fused kernel, shows no such pathology. The fused kernel of §6 eliminates the entire allocation chain. **The kernel rewrite is therefore a structural prerequisite for production deployment, not an optimisation.**

Table 5: Multi-batch profiling at $N = 5\text{K}$, RTX 4060 Ti (16 GB).

Path	Batch	lat/step	lat/elem	GPU util	peak VRAM
Python DCR	1	43.0 ms	43.0 ms	∼57%	—
Python DCR	4	54.8 ms	13.7 ms	96%	3.48 GB
Python DCR	8	OOM thrash	>60 (deg.)	100% thrash	16.0 GB
SDPA	1	30.4 ms	30.4 ms	63%	2.69 GB
SDPA	4	∼35 ms	8.8 ms	91%	—

7 Methodology: matched-control discipline

The quantitative claims above are validated under a discipline we make explicit, because it is the discipline — not any single number — that lets the claims survive. It originates in a worked *null* from a closed companion programme.

A worked example (the closed spectral programme). The free-energy Hessian of an attention head can be written $H = -\beta\,\mathrm{Cov}_p(k)$, and on trained models it is typically spike-dominated. The hypothesis under test was that this dominant mode is a privileged computational object whose removal should selectively damage next-token prediction. The natural intervention — project out the dominant direction, then rescale to preserve $\|K\|_F$ — preserves the *wrong* invariant. Removing the dominant direction deletes a large key component; removing a bulk direction deletes a tiny one; rescaling to a common $\|K\|_F$ leaves a perturbation-norm asymmetry $\|\delta K\|_F$ of 10–30× (for one early-layer head, $M_T \approx 84.9$ vs. $M_C \approx 15.8$). An uncontrolled smoke run accordingly showed the treatment producing $\sim 18\times$ the control's KL — "a striking positive," and entirely consistent with the dominant mode being computationally irrelevant. The asymmetry was caught not by inspecting the result but by directly measuring $\|\delta K\|_F$ before the full run.

The three-component checklist.

1. **Matched-magnitude intervention.** Match the *perturbation norm* $\|\delta\|$ applied to the model — not an indirect invariant such as $\|K\|_F$. We scaled both treatment and control to $T = \min(M_T, M_C)$ per layer–head pair; the verified treatment-to-control ratio was 1.000 on every pair.
2. **A random-spectrum baseline.** Before attributing significance to a spectral object, check whether a random spectrum exhibits the same property. A three-statistic coupling that looked like trained structure was reproduced exactly by random spectra — an algebraic identity among monotone functions of spectral peakedness, not a property of the network.
3. **Pre-registered outcome criteria.** Fix, before running, what each outcome means, so a null is as reportable as a positive and there is no post-hoc freedom to reinterpret.

Result of applying the framework. On Llama-3.2-1B over WikiText-2 (twelve layer–head pairs, 25 prompts each, 300 measurements, context 256), the matched-control ablation returned a pre-registered *null* (Table 6): the treatment-minus-control next-token NLL differential is $+2.72 \times 10^{-4}$, 95% CI $[-1.7, 7.0] \times 10^{-4}$ bracketing zero, Wilcoxon $p = 0.26$. Suppressing the dominant spectral mode harms prediction no more than suppressing a random mode of equal perturbation magnitude. Note that the *KL* differential *is* significant ($p < 10^{-38}$) — a dissociation that itself argues for choosing the operationally meaningful metric (NLL) before running.

Table 6: Matched-control ablation of the dominant spectral mode. NLL — the operationally meaningful metric — gives a differential whose 95% CI brackets zero.

Metric	Δ_{treat}	Δ_{ctrl}	differential	Wilcoxon p
NLL (next-token)	5.95×10^{-4}	3.23×10^{-4}	$+2.72 \times 10^{-4}$	0.26
KL (distribution)	1.31×10^{-3}	7.81×10^{-4}	$+5.32 \times 10^{-4}$	$< 10^{-38}$

Why this is the spine. The lesson is reproducible, not philosophical: the same attention geometry, on the same model and dataset, was *falsified* as a mechanism and *survived* as a description and as algebra. We therefore apply the checklist as the validation standard for every quantitative claim in this paper, and we state in §10 the pre-registered matched-control test that would apply it to Theorem 2 itself.

8 Distinction from concurrent work

The mainstream sparse-attention literature converges on "token-relevance indexer + top-k selection + sparse downstream attention," differing in (a) whether the indexer is trained or training-free and (b) whether the criterion has formal guarantees. Ours is the only training-free path with an *exact* ranking guarantee (Theorem 1) plus a *closed-form scaling law* (Theorem 3) and a *per-query certificate* (§3.3). Theorem 1 is a special case of the MIPS-as-query-scaled-NNS equivalence with $u_Q = Q/\|Q\|_2$ specific to attention. Concurrent theory derives a TV-distance bound for top-k truncation (which we use) and observes qualitatively that low-entropy states suit top-k decoding, but does not derive a scaling law in N; our measured exponent α and the rate $O(N^{-(1-\alpha)})$ are the new content. We note the precedence pressure honestly: the qualitative entropy$\leftrightarrow$top-k observation is already public, which is part of why we release for priority now and scope the scaling claim as descriptive pending the causal test.

9 Limitations and retraction ledger

Limitations. Single model family (Llama-3.2-1B) and single dataset (WikiText-2); α may differ across architectures and domains. Triton/ABKV figures are partly projected from measured anchors; the memory-wall finding is preliminary and requires the kernel rewrite to enable batched validation. Retention claims are at fixed $k_{\text{window}} = 64$ on one base; multihop floors on the 1B base. KV-cache memory is not addressed (orthogonal and complementary to Q-axis).

Retraction ledger. In the interest of an honest permanent record, we list hypotheses entertained in the programme and *retracted*, with how each fell:

R1 Spectral geometry as a computational *mechanism* (dominant mode load-bearing) — matched-magnitude null (§7).

R2 A universal U-shaped depth profile of spectral concentration — did not reproduce across five architectures; model-specific.

R3 A non-trivial three-statistic spectral coupling — reproduced by random spectra; algebraic identity.

R4 Tail-equivalence $\psi(\alpha, k) \to \tau_k^2$ and the closed form $\alpha^* = r - \frac{1}{2}$ — not predictive at operational k (§3.3).

R5 An absolute-k_{eff} retention threshold ("K_{crit}") — $k_{\text{eff}} = 19{,}666$ fails while $k_{\text{eff}} = 1{,}602$ holds; the boundary is an $N\times$density interaction.

R6 Retention failure mode as mis-selection — inverted by the data; starvation is the only currently significant mechanism (pending inject, §5).

R7 Super-linear Python overhead as the memory-wall mechanism — the mechanism is allocator fragmentation (§6).

R8 Pass-3 as the kernel bottleneck — Pass-2 dominates (74.4% of M4, Table 4).

10 Conclusion and future work

We assembled, around the single object of softmax concentration, an exact ranking identity (Theorem 1), a measured and descriptively-stated scaling law (Theorem 3) with two independent confirmations, a pre-registered retention characterisation with a clean leakage dissociation, a runtime-checkable per-query certificate, and a systems result in which a fused kernel is a structural prerequisite. Every claim was held to a matched-control discipline born from a falsified spectral hypothesis.

Pre-registered next steps (not yet claimed).

- **Gap-1 — causal upgrade of Theorem 3.** A matched-magnitude causal test of the entropy-concentration premise; the protocol and both pre-registered outcomes are specified below.
- **Retention gates.** Multi-seed the 128K boundary; settle starvation vs. mis-selection by oracle-inject; validate the needle-in-top-k metric by anti-inject; decompose latency into TTFT (dense prefill, not accelerated) and per-token decode.
- **Cross-model α.** Measure α across softmax-scaling regimes (vanilla+RoPE, SSMax, YaRN, α-entmax) to convert "model-specific" from a caveat into a law.
- **Low-rank Q-axis (DCR-Loki).** Apply Theorem 1 in the top-r PCA subspace of K; reducing $d = 128 \to r = 8$ targets a $\sim 16\times$ data-movement reduction on top of the 50% compute saving, with a TV-distance error budget.

Registered protocol (Gap-1), recorded as of this release. We register the causal test of Theorem 2 before running it, so that this release stakes the test design. Object: the premise that entropy concentration is causally load-bearing. For per-query logits $\boldsymbol{\ell} = QK^\top/\sqrt{d}$, $p = \mathrm{softmax}(\boldsymbol{\ell})$, $H(p) = -\sum_i p_i \log p_i$, with entropy gradient $g = \nabla_{\boldsymbol{\ell}} H = -p \odot (\log p + H)$:

- **Treatment** δ_T: a step along $+g/\|g\|$ sized so the realised $\Delta H \approx \Delta_{\text{target}}$ nats (destroying concentration toward uniform).
- **Matched control** δ_C: a random direction projected onto $\{v : \langle v, \hat{g}\rangle = 0\}$ and scaled to $\|\delta_C\| = \|\delta_T\|$, so H is preserved to first order. The matched invariant is the *perturbation norm* $\|\delta\boldsymbol{\ell}\|$ — not $\|\boldsymbol{\ell}\|$ — which is precisely the invariant the spectral-mechanism study of §7 got wrong before correction.
- **Contract gates** (else a cell is invalid): verified ratio $\|\delta_T\|/\|\delta_C\| = 1.000 \pm 10^{-3}$ and measured $|\Delta H_C| \le 0.1\,\Delta_{\text{target}}$.
- **Random-K baseline**: whether untrained K yields the same α, contextualising algebraic vs. trained origin.
- **Primary metric**: paired next-token NLL differential $D_{\text{NLL}} = \text{NLL}_{\text{treat}} - \text{NLL}_{\text{ctrl}}$; statistics by 10k-resample bootstrap 95% CI and Wilcoxon signed-rank, mirroring Table 6; design 12 layer–head pairs $\times$ 25 prompts, multi-seed before any final claim.

Pre-registered outcomes (fixed now). If the 95% CI of D_{NLL} brackets zero *and* Wilcoxon $p > 0.05$: Theorem 2 is finalised as *descriptive* under a matched-control null. If the CI is strictly positive *and* $p < 0.05$: the concentration premise is *causally grounded.* No reinterpretation of a null as a weak positive is permitted; the decision is read off the CI and Wilcoxon test. The result will be published as version `v1.1` (versioned DOI; concept DOI unchanged), by uncommenting the corresponding branch already prepared in §3.

A Proof of Theorem 1

For any $Q \neq 0$, $\langle Q, K\rangle = \|Q\|_2 \langle u_Q, K\rangle$ with $u_Q = Q/\|Q\|_2$ and $\|Q\|_2 > 0$. Hence for any i, j, $\langle Q, K_i\rangle - \langle Q, K_j\rangle = \|Q\|_2(\langle u_Q, K_i\rangle - \langle u_Q, K_j\rangle)$, and the sign is preserved because $\|Q\|_2 > 0$. The softmax is strictly monotone in its logits, so the score ordering equals the projection ordering exactly. Ties of measure zero in exact arithmetic appear at finite rate ($\sim$ 12% z-ties) under `bf16` and are resolved by a stable tie-break (smaller index wins), which is the only source of the $\sim 10^{-3}$ residual in the 45/45 brute-force check (§4).

B Proof sketch of Theorem 3, and the kernel plan

Proof sketch. The coverage floor admits $k_{\text{eff}} = \lfloor c_{\text{floor}} N \rfloor$ keys. Under Assumption 1, $K_{\text{eff}}(N) \sim N^{\alpha}$, so $k_{\text{eff}}/K_{\text{eff}} \sim N^{1-\alpha} \to \infty$. By Theorem 1 the selected window is exactly the top-k_{eff} subset by attention score, so the captured probability mass $\to 1$. By the head–tail decomposition, $\|\text{Attn} - \text{Attn}_{k_{\text{eff}}}\|_2 = \tau_N \|\mu_{\text{tail}} - \mu_{\text{head}}\|_2$ with $\tau_N = \text{TV}(P_N, \hat{P}_N) = O(N^{-(1-\alpha)})$; translating the output perturbation to perplexity via the cross-entropy expansion preserves the rate. The worst-head exponent α^* controls the rate; head-averaged α lower-bounds it.

Kernel engineering plan (M0–M7). A FlashDecoding-style kernel: KV in HBM; decode streams over KV in 64-key blocks, projecting onto u_Q and maintaining a running top-k via warp-shuffle merge; selected $K_{\text{sel}}, V_{\text{sel}}$ pass through a fused online-softmax output kernel. The Pareto-optimal target tile regime is $k_{\text{eff}} \in [0.3N, 0.5N]$ (§4). The ABKV variant replaces decode-time selection with a prefill prefix sort (§6); a Triton bitonic top-k (M6) is retained as a fallback if the prefix mechanism fails to transfer from synthetic to real attention distributions.

C Production artifact: sort library and explainable dispatch

The framework ships as a C++17/AVX-2 sort library (validated to 21.6× over `std::sort` on AVX-2, with bit-exact gradient verification $\|\nabla\|_2 = \sqrt{n}$ to machine precision and 36/36 correctness targets) and an AVX-512 build stress-tested at $\approx$ 2,800 correctness operations (one performance gate failure, no correctness failures), a 528-case adversarial sweep with zero failures, byte-identical determinism over 50 repeats, and linear scaling validated to 64M elements. The runtime exposes an *explainable* dispatch surface: the strategy, inversion density, NaN flag, and threads used per call are queryable, so adversarial tests can assert *which* path ran, not merely that the output was correct — "runtime decisions become falsifiable." This is the same observe-before-decide discipline as §7, realised in software.

D Matched-control raw and the Detector v2 dispatcher

The matched-control run is the 300-measurement design of §7: twelve layer–head pairs, 25 prompts, context 256, three forward passes per measurement (unablated, dominant-mode treatment, matched bulk-mode control). Detector v2 partitions queries by a coherence score s_{v2} and an axis confidence c into Dense / Rank-local / Hybrid regimes; the no-false-negative guarantee (Lemma 6.1 of the companion framework) is realised empirically as 15/15 random queries routed to dense and zero false negatives over the 45 calibration cells of §3.3. The multi-axis extension predicts the axis count $r^\star = \lceil 1.142\, d_{\mathrm{eff}}(Q) - 1.048 \rceil$ (Pearson 0.9925 over 40 trials, 32–86% relative error reduction over single-axis), promoting $d_{\mathrm{eff}}(Q)$ from a regime selector to a quantitative axis-count predictor and matching the structural negative result (Proposition 5.13: sparsity in the query spectrum does not imply rank-local recoverability under single-axis PCA; realised as a controlled -17 pt BERT-MNLI drop).

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